

Experiment 29

Aim

To perform morphological erosion on an image using OpenCV.

Procedure

1. Import the required libraries, OpenCV and NumPy.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a morphological kernel using NumPy.
5. Apply the Erosion operation using cv2.erode().
6. Display the original and eroded images.
7. Save the resulting image.

Code:

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found!")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((5, 5), np.uint8)
eroded = cv2.erode(gray, kernel, iterations=1)
cv2.imshow("Original Image", img)
cv2.imshow("Eroded Image", eroded)
cv2.imwrite("eroded_image.jpg", eroded)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Result

The morphological erosion operation was successfully performed using OpenCV. The erosion process reduced the boundaries and small white regions in the image, and the resulting image was saved as `eroded_image.jpg`.